

Mathematics A

Unit 1 • Chapter OL-T50 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

4 classified items • 19 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

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Contents

Unit 1 • Expertise • Unit 1 • Chapter OL-T50 • Expertise Q16+ • 4 items • 19 marks

Chapter	Items	Marks	Starts
01. OL-T50 3D Pythagoras & Trigonometry	4	19	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

3D Pythagoras & Trigonometry

Unit 1 • Expertise • 19 marks • 4 item(s)

January 2022 | 4MA1-1H Q21 | 5 marks

Use midpoint or symmetry structure in a 3D model

June 2021 | 4MA1-1H Q22 | 4 marks

Use a non-rectangular base before a line-plane angle

May-Nov 2020 | 2H Q18 | 5 marks

Find angle between a line and a plane

January 2023 | 2H Q23 | 5 marks

Use 3D trigonometry to obtain a terminal volume

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-1H Q21

5 marks

- 21 The diagram shows the prism $ABCDEFGHJK$ with horizontal base $AEFG$

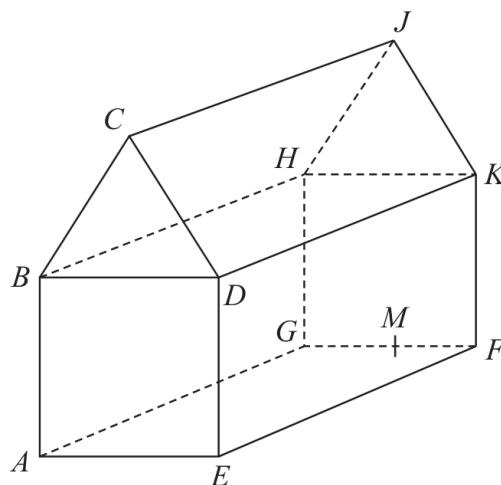


Diagram **NOT**
accurately drawn

$ABCDE$ is a cross section of the prism where
 $ABDE$ is a square
 BCD is an equilateral triangle

$$EF = 2 \times AE$$

M is the midpoint of GF so that JM is vertical.

$$\text{Angle } MAJ = y^\circ$$

$$\text{Given that } \tan y^\circ = T$$

find the value of T , giving your answer in the form $\frac{\sqrt{p} + \sqrt{q}}{17}$ where p and q are integers.

$T = \dots\dots\dots$

Continue your method**5 marks**

WORKING SPACE

22 The diagram shows a triangular prism $ABCDEF$ with a horizontal base $ABEF$.

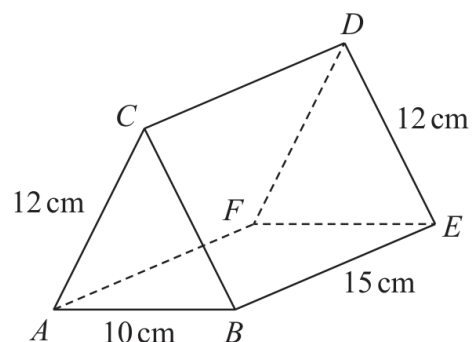


Diagram **NOT**
accurately drawn

$$AC = BC = FD = ED = 12 \text{ cm} \quad AB = 10 \text{ cm} \quad BE = 15 \text{ cm}$$

Calculate the size of the angle between AD and the base $ABEF$.
Give your answer correct to 3 significant figures.

YOUR WORKING

18 The diagram shows cuboid $ABCDEFGH$.

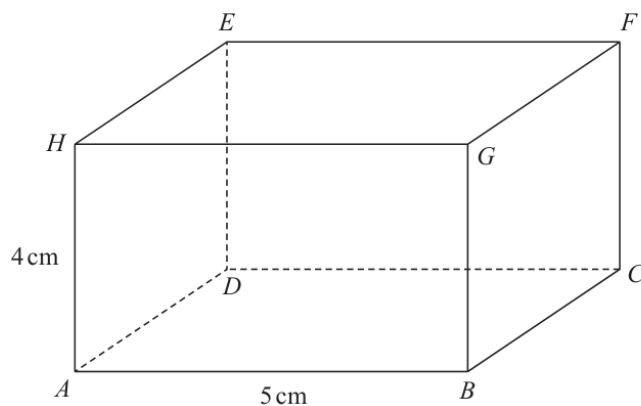


Diagram **NOT**
accurately drawn

$$AB = 5 \text{ cm}$$

$$AH = 4 \text{ cm}$$

The size of the angle between CH and the plane $ABCD$ is 35°

Calculate the volume of the cuboid.

Give your answer correct to 3 significant figures.

.....cm³

(Total for Question 18 is 5 marks)

YOUR WORKING

23 The diagram shows a solid prism $ABCDEFGHJ$

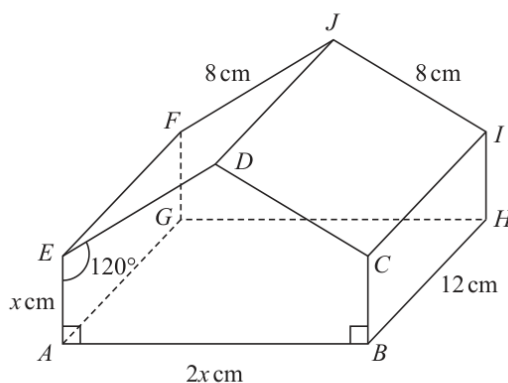


Diagram **NOT**
accurately drawn

The prism is such that each cross section is a pentagon where

$$AE = BC = x \text{ cm} \quad AB = 2x \text{ cm} \quad ED = CD = 8 \text{ cm}$$

$$\text{angle } EAB = \text{angle } CBA = 90^\circ \quad \text{angle } AED = \text{angle } BCD = 120^\circ$$

Given that $AG = BH = EF = DJ = CI = 12 \text{ cm}$

calculate the angle that AJ makes with the base $ABHG$ of the prism.

Give your answer correct to 3 significant figures.



YOUR WORKING

(Total for Question 23 is 5 marks)

YOUR WORKING